

Ques. WHAT ARE MEMS SENSORS? WHAT ARE THE TYPES AVAILABLE IN THE MARKET?

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Ans. Microelectromechanical systems (MEMS) are a class of devices characterised both by their small size and the manner in which these are made. These are made up of component sizes between 1 and 100 micrometres. MEMS devices can vary from simple structures to extremely complex electromechanical

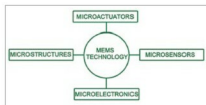


Fig. 1: General block diagram of a MEMS device

systems with multiple moving elements under the control of integrated micro-electronics.

Generally, MEMS consist of mechanical microstructures, microactuators, microensors and microelectronics in one package. Fig. 1 shows the block diagram of MEMS device.

Some common types of MEMS sensors available in the market are:

MEMS accelerometers. These sensors are used in smartphones for various controls including switching between landscape and portrait modes, anti-blur capture and pocket-mode operation.

MEMS gyroscopes. These detect the angular displacement of an object. MEMS gyros are used for vehicle stability control.

MEMS pressure sensors. These sensors are used in automotive, industrial, medical, defence and aerospace applications.

MEMS magnetic field sensors. These sensors find use in position sensing, current detection, speed detection,

vehicle detection, space exploration, etc.

Fluxgate sensors. These find many applications in space research, geophysics, mineral prospecting, automation and industrial process control.

Q2. WHAT TYPES OF WIRELESS COMMUNICATION TECHNOLOGY CAN BE USED IN HOME AUTOMATION?

Pournima

A2. In wireless communication, signals are transmitted through air without using any cable, wires or other electrical conductors. Wireless communication technologies include infrared (IR), mobile phone, radio frequency (RF), Wi-Fi, Bluetooth, Li-Fi, ZigBee, Z-Wave, laser, ultrasonic, etc. These find many applications in diverse fields. Home automation is one of them. Typically, a wireless home automation system comprises devices as shown in Fig. 2.

Some wireless communication devices used in home automation are:

Infrared. Here, information is shared between devices or systems through IR radiation. In the electromagnetic spectrum, IR radiation lies between microwaves and visible light. It is used in many consumer electronics including security system, TV remote control and short-range communications.

Mobile phone. It can be used to turn on/off various electrical appliances

es in homes. So, it works as a remote for home automation system. Most control signals from the mobile phone are sent either using Bluetooth or GSM (Global System for Mobile communication) mode in India. In GSM mode, it is possible to control home appliances from very far off as cellular networks have a wide range of coverage.

Wi-Fi. Appliances can be controlled using a personal computer through Wi-Fi. Most Wi-Fi devices use 2.4GHz frequency and implement frequency-division multiplexing technology.

Bluetooth. Bluetooth devices are easily accessible for wireless control as these exist in many electronic products including mobile phones. Bluetooth provides a range of about 10 metres at the speed of 1Mbps.

Radio frequency. You can control home appliances using a pair of RF modules (transmitter and receiver). For example, most common RF devices used by hobbyists are 433MHz RF modules. ZigBee is a more powerful and longer-range RF device.

ZigBee. It is a low-power, wireless mesh network that operates in the following radio bands: 868MHz, 915MHz, and 2.4GHz.

Z-Wave. It is a relatively new wireless home automation protocol. It uses extremely less power and runs on a mesh network. The device communicates in a sub-gigahertz frequency range, around 900MHz.

Li-Fi. Light Fidelity or Li-Fi uses the visible region of the electromagnetic spectrum for communication. It can be easily used in places where Bluetooth, IR and Wi-Fi are banned, because it uses light instead of radio waves.



Fig. 2: Typical wireless home automation system

Answers compiled by Sani Theo, technical editor.
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